

Dynamic Multi-Teacher Knowledge Distillation for Efficient LLM Compression

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Abstract

State-of-the-art Large Language Models (LLMs) possess remarkable reasoning capabilities but suffer from high computational costs, making deployment on edge devices challenging. This research investigates **Knowledge Distillation (KD)** strategies to compress massive, arbitrary “Teacher” models into a compact ($\approx 250\text{M}$ parameter) “Student” model. We contrast two distinct methodologies: **Snapshot Distillation (Offline)**, where teacher outputs are pre-computed and stored, and **Direct Distillation (Online)**, where an ensemble of teachers generates supervision signals in real-time. Our proposed architecture leverages modern Transformer components—including Grouped Query Attention (GQA), Rotary Embeddings (RoPE), and SwiGLU activations—to maximize the student’s ability to retain knowledge from any combination of teacher models.

1 Introduction

The democratization of Artificial Intelligence is currently hindered by the hardware requirements of high-performance LLMs. While techniques like quantization reduce the memory footprint, they do not fundamentally change the model’s architecture. Knowledge Distillation offers a solution by training a smaller network (Student) to mimic the probability distributions of larger networks (Teachers).

The core hypothesis of this project is that an ensemble of diverse teachers provides a richer training signal than a single teacher. This system is designed to be **model-agnostic**, allowing researchers to plug in *any* available LLMs as teachers. This paper details the architecture of a custom Student LLM and evaluates the trade-offs between Snapshot and Direct distillation methodologies.

2 Methodology: Distillation Strategies

We explore two fundamental approaches to transferring knowledge from arbitrary Teacher models to the Student.

2.1 Method A: Snapshot Distillation (Offline)

In this traditional approach, the distillation process is decoupled into two phases:

1. **Generation Phase:** Teacher models iterate through the training dataset. Raw logits are extracted and saved to disk.
2. **Training Phase:** The Student model is trained using the dataset and saved logits without loading Teachers into memory.

While memory-efficient, this method incurs massive storage costs and prevents dynamic data augmentation.

2.2 Method B: Direct Multi-Teacher Distillation (Online)

This project implements an online approach where Teachers and Student reside in GPU memory simultaneously.

The Ensemble Process: For every training step with input batch x , the ensemble target $Z_{ensemble}$ is computed dynamically:

$$Z_{ensemble} = \sum_{k=1}^N \alpha_k \cdot Z_{teacher_k}(x) \quad (1)$$

where α_k represents the weight assigned to Teacher k . This allows for real-time data augmentation and instant swapping of teacher models.

3 Student Model Architecture

The Student model is designed to be parameter-efficient ($\approx 250M$ parameters) while incorporating architectural advancements found in modern SOTA LLMs.

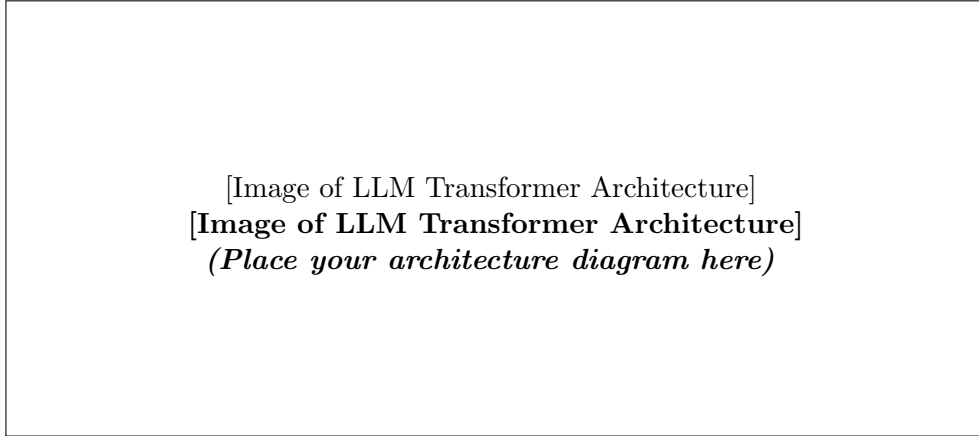


Figure 1: The Student LLM Architecture featuring RMSNorm, SwiGLU, and GQA.

3.1 Root Mean Square Normalization (RMSNorm)

We replace LayerNorm with RMSNorm to stabilize gradients. RMSNorm normalizes inputs based on the root mean square of activations:

$$\text{RMS}(x) = \sqrt{\frac{1}{d} \sum_{i=1}^d x_i^2 + \epsilon} \quad (2)$$

The transformation is then applied as:

$$\bar{x}_i = \frac{x_i}{\text{RMS}(x)} \cdot \gamma_i \quad (3)$$

3.2 SwiGLU Activation

The Feed-Forward Network utilizes the SwiGLU activation, which combines the Swish function with a Gated Linear Unit for improved gradient flow:

$$\text{SwiGLU}(x) = (\text{Swish}(xW_g) \otimes (xW_u)) W_d \quad (4)$$

where $\text{Swish}(x) = x \cdot \sigma(x)$.

3.3 Rotary Positional Embeddings (RoPE)

RoPE encodes position by rotating the embedding vectors in the complex plane, ensuring attention scores depend only on relative distance:

$$\begin{pmatrix} x'_1 \\ x'_2 \end{pmatrix} = \begin{pmatrix} \cos m\theta & -\sin m\theta \\ \sin m\theta & \cos m\theta \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \quad (5)$$

3.4 Grouped Query Attention (GQA)

To reduce memory bandwidth during inference, we utilize GQA. Multiple Query heads (H_q) share a single Key/Value head (H_{kv}), where the replication factor is $R = H_q/H_{kv}$.

4 Loss Function & Optimization

The training objective combines soft targets (from Teachers) and hard targets (Ground Truth).

4.1 Composite Loss Function

The total loss $\mathcal{L}_{total}$ is calculated as:

$$\mathcal{L}_{total} = \alpha \cdot \mathcal{L}_{KL} + (1 - \alpha) \cdot \mathcal{L}_{CE} \quad (6)$$

1. KL Divergence (Soft Target): We minimize the divergence between the Student’s distribution Q and the Teacher’s ensemble distribution P , scaled by temperature T :

$$\mathcal{L}_{KL} = T^2 \sum_x P(x) \log \left(\frac{P(x)}{Q(x)} \right) \quad (7)$$

2. Cross-Entropy (Hard Target): Standard next-token prediction loss against the actual dataset labels:

$$\mathcal{L}_{CE} = - \sum_i y_i \log(p_i) \quad (8)$$

4.2 Optimization Setup

- **Optimizer:** AdamW with decoupled weight decay ($\lambda = 0.01$).
- **Scheduler:** Cosine Annealing with Linear Warmup.
- **Precision:** Mixed Precision (BF16/FP16) for memory efficiency.

5 Conclusion

This framework provides a flexible, high-performance pathway for distilling state-of-the-art reasoning capabilities into efficient, deployable models. By utilizing direct multi-teacher distillation, we enable dynamic curriculum learning and data augmentation that static snapshot methods cannot achieve.